

Function Point Analysis (FPA)

A Concise Mathematical & Calculation Reference Guide

Functional Point Analysis (FPA) is an objective, technology-independent software measurement technique used to estimate the size, complexity, and development effort of a system based entirely on the functional requirements delivered to the user.

1. The Five Core Functional Components

FPA categorizes software requirements into two primary functional types, split into five specific information domain characteristics:

A. Transactional Function Types

- **External Inputs (EI):** Processes data/control updates arriving from outside the system boundary (e.g., input forms, add/delete actions).
- **External Outputs (EO):** Processes data leaving the boundary that includes derived/calculated logic (e.g., reports, calculated charts).
- **External Inquiries (EQ):** Direct data retrieval and simple output generation mapping directly from sources without derived or calculated fields.

B. Data Function Types

- **Internal Logical Files (ILF):** Logically related groups of user-identifiable data completely maintained *inside* the system boundary (e.g., internal database tables).
- **External Interface Files (EIF):** User-identifiable reference data referenced by the system but maintained entirely *outside* the boundary by another system.

2. Standard Functional Weight Matrix

Each information domain characteristic is assigned an empirical weighting factor based on its functional complexity (Low, Average, High):

Measurement Parameter (Functional Unit)	Low Complexity	Average Complexity	High Complexity
External Input (EI)	3	4	6
External Output (EO)	4	5	7
External Inquiry (EQ)	3	4	6
Internal Logical File (ILF)	7	10	15
External Interface File (EIF)	5	7	10

3. Mathematical Formulation & Calculation Steps

Core Equation:

$$FP = UFP \times CAF$$

Where:

- **UFP** = Unadjusted Function Points = $\sum (\text{Count of item} \times \text{Weighting Factor})$
- **CAF** = Complexity Adjustment Factor = $0.65 + [0.01 \times \sum(f_i)]$
- $\sum(f_i)$ = Total Degree of Influence (DI) across 14 General System Characteristics (GSCs)

1. **Compute UFP:** Classify each component (EI, EO, EQ, ILF, EIF) by complexity level, multiply its counts by the respective weights from the matrix, and sum them up.
2. **Evaluate Complexity Scale:** Assess 14 standard questions from a scale of **0 to 5**:
0: No Influence | 1: Incidental | 2: Moderate | 3: Average | 4: Significant | 5: Essential
Sum these values up to obtain $\sum(f_i)$, which will fall between 0 and 70.
3. **Compute CAF:** Plug the sum into the formula. Note that **CAF** structurally ranges from **0.65** (when DI=0) to **1.35** (when DI=70).
4. **Final Calculation:** Multiply **UFP** $\times$ **CAF** to get the total system Function Points.

4. Solved Practical Examples

Example 1: Core Standard Calculation (Average Complexities)

Scenario: A project metrics evaluation has yielded the following counts: 50 User Inputs, 40 User Outputs, 35 User Inquiries, 6 Internal Files, and 4 External Interfaces. All weighting factors and the complexity adjustment factors are determined to be **average**. Compute the final Function Points.

Solution:

Step 1: Calculate Unadjusted Function Points (UFP)

Using the *Average* column values from the weights table (EI=4, EO=5, EQ=4, ILF=10, EIF=7):

$$\text{UFP} = (50 \times 4) + (40 \times 5) + (35 \times 4) + (6 \times 10) + (4 \times 7)$$

$$\text{UFP} = 200 + 200 + 140 + 60 + 28 = 628$$

Step 2 & 3: Find Degree of Influence & CAF

Since the adjustment factors are stated to be "average", the scale value for all 14 factors is 3.

$$\sum(f_i) = 14 \times 3 = 42$$

$$\text{CAF} = 0.65 + (0.01 \times 42) = 0.65 + 0.42 = 1.07$$

Step 4: Compute Final Function Points (FP)

$$\text{FP} = \text{UFP} \times \text{CAF}$$

$$\text{FP} = 628 \times 1.07 = 671.96$$

Final Answer: 671.96 Function Points

Example 2: Advanced Variable Adjustments (GATE Examination Level)

Scenario: A software project presents the following specifications:

- Characteristics: External Inputs = 30, External Outputs = 60, External Inquiries = 23, Internal Files = 8, External Interfaces = 2.
- Explicitly Given Complexity Weights: EI=4, EO=5, EQ=4, ILF=10, EIF=7.
- Value Adjustment Factors: Out of 14 factors, 4 are completely not applicable (value 0), 4 factors have a value of 3, and the remaining 6 factors have a value of 4. Calculate the total Function Points.

Solution:

Step 1: Calculate Unadjusted Function Points (UFP)

Multiply the specific component counts by their designated explicit weights:

$$\text{UFP} = (30 \times 4) + (60 \times 5) + (23 \times 4) + (8 \times 10) + (2 \times 7)$$

$$\text{UFP} = 120 + 300 + 92 + 80 + 14 = 606$$

Step 2 & 3: Find $\sum(f_i)$ and Complexity Adjustment Factor (CAF)

Break down the 14 adjustment factors to compute the total sum:

$$\sum(f_i) = (4 \times 0) + (4 \times 3) + (6 \times 4)$$

$$\sum(f_i) = 0 + 12 + 24 = 36$$

$$\text{CAF} = 0.65 + (0.01 \times 36) = 0.65 + 0.36 = 1.01$$

Step 4: Compute Final Function Points (FP)

$$\text{FP} = \text{UFP} \times \text{CAF}$$

$$\text{FP} = 606 \times 1.01 = 612.06$$

Final Answer: 612.06 Function Points